

The potential and limitations of neutrons, electrons and X-rays for atomic resolution microscopy of unstained biological molecules

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I. SUMMARY

Radiation damage is the main problem which prevents the determination of the structure of a single biological macromolecule at atomic resolution using any kind of microscopy. This is true whether neutrons, electrons or X-rays are used as the illumination. For *neutrons*, the cross-section for nuclear capture and the associated energy deposition and radiation damage could be reduced by using samples that are fully deuterated and ^{15}N -labelled and by using fast neutrons, but single molecule biological microscopy is still not feasible. For naturally occurring

biological material, *electrons* at present provide the most information for a given amount of radiation damage. Using phase contrast electron microscopy on biological molecules and macromolecular assemblies of $\sim 10^5$ molecular weight and above, there is in theory enough information present in the image to allow determination of the position and orientation of individual particles: the application of averaging methods can then be used to provide an atomic resolution structure. The images of approximately 10000 particles are required. Below 10^5 molecular weight, some kind of crystal or other geometrically ordered aggregate is necessary to provide a sufficiently high combined molecular weight to allow for the alignment. In practice, the present quality of the best images still falls short of that attainable in theory and this means that a greater number of particles must be averaged and that the molecular weight limitation is somewhat larger than the predicted limit. For *X-rays*, the amount of damage per useful elastic scattering event is several hundred times greater than for electrons at all wavelengths and energies and therefore the requirements on specimen size and number of particles are correspondingly larger. Because of the lack of sufficiently bright neutron sources in the foreseeable future, electron microscopy in practice provides the greatest potential for immediate progress.

2. INTRODUCTION

It would be extremely valuable if the magnification of an optical microscope could simply be increased to atomic resolution by clicking in higher powered objective lenses. It would then be possible to start with an overall view of a living cell, close in on sub-cellular structures and end with atomic resolution close-ups of single biological macromolecules. Unfortunately, normal optical microscopy is limited to resolutions of about 200 nm by the wavelength of light. The attempt, in biology, to use shorter wavelengths creates the immediate problem of U.V. radiation damage, which becomes more severe as the wavelength of the electromagnetic radiation is further shortened into the high resolution X-ray region. Traditionally in biology, microscopy in the resolution range from 200 nm down to 0.2 nm has been the realm of the electron microscope. Here too, radiation damage is a major limitation for work on unstained biological specimens. Finally, although neutron microscopy at higher resolution has never really been a practical proposition because neutron fluxes have been much too low, radiation damage through nuclear reactions is also a major limitation.

With progress being made in all three of these areas, critical comparisons are needed from time to time. What follows is an attempt to review the current situation. Many of the points are obvious and widely known, but in recent discussions, it has become clear to me that many simple facts considered self-evident in one area are not widely appreciated in another. I hope that this synopsis will therefore help to establish some common ground. The review does not include any consideration of near-field microscopy using scanning probes of any kind, nor methods which only look at surfaces. Previous publications which have considered the relative merits of neutron, X-ray or electron microscopy include

Breedlove & Trammel (1970), Hoppe (1983) and Sayre *et al.* (1977). Useful text books include Cowley (1975), Reimer (1989) and Spence (1988).

3. PHASE CONTRAST VERSUS OTHER MODES OF MICROSCOPY

In phase contrast microscopy, illumination which is coherently scattered by the object is focused back by the objective lens to interfere coherently with the transmitted beam. In optical phase contrast microscopy with the Zernike quarter wave plate, the 90° phase shift on scattering is augmented by a 90° phase shift applied to the unscattered beam so that the scattered and unscattered beams interfere to produce the maximum contrast. The quarter wave plate is often also given a finite neutral density so that the direct beam is reduced in intensity (Zernike, 1955). This gives greater relative contrast and makes it easier for the eye to see details. However it is easy to show that for a perfect detector this reduction in the intensity of the unscattered beam has no effect on the signal to noise ratio of features in the resulting image for the same incident flux of quanta.

In bright-field phase contrast electron microscopy, the corresponding relative phase shifts of the diffracted beams are normally produced by defocus and spherical aberration (Thon, 1966; Erickson & Klug, 1971). Although the effect on the image is more complex than with the Zernike in-focus method, the resulting peak contrast levels and signal-to-noise ratios are identical at spatial frequencies corresponding to maxima in the contrast transfer function, and interpretable images are easily produced by digital image processing.

In weak phase contrast where the object scatters only a small proportion of the incident illumination, the image contains a signal proportional to the amplitude of the scattering, and is linearly related to properties of the object. Each scattered quantum, via interference between its wavefront and that of the unscattered beam, produces a signal in the image which is retrievable with the same signal-to-noise ratio on amplitude as from counting statistics (Henderson, 1992). The phase is also determined in the image with the same signal-to-noise ratio as the amplitude.

By comparison, other modes of microscopy which are based on absorption of the incident beam or on analysis of scattered or absorbed quanta in scanning modes, rely on positive or negative signals that arise only from counting quanta. In these other modes, the signal is proportional to the intensity of the scattered, absorbed or otherwise perturbed incident beam. Because of this, for all weakly interacting materials, when the signal is measured per absorbed or scattered quantum the phase contrast mechanism is always more efficient at producing a signal such as a feature in the image – if the amplitude of a scattered beam is a fraction, f , of the incident beam, phase contrast gives a signal proportional to f whereas amplitude contrast is proportional to f^2 . A good example of this comes from the work of Smith & Langmore (1992). They have recorded images of tobacco mosaic virus (TMV) in ice where the ratio of elastic scattering to inelastic scattering is approximately 1:3. Elastic scattering is usable for phase contrast whereas inelastic scattering is usable for amplitude contrast via energy filtering to remove energy loss electrons. Amplitude contrast is here defined as all contrast

due to lost electrons, usually understood as absorption and aperture contrast by the electron microscopy community. The proportion of inelastic electron scattering is higher if resolutions less than $\frac{1}{2}\text{\AA}^{-1}$ are considered (Reimer, 1989, p. 148). The contrast Smith & Langmore find due to absorption (produced in part by energy filtering) is only 14% of that due to phase contrast. At first sight, this suggests, for that particular sample and at that electron energy (80 keV), that phase contrast is about 20 (i.e. $\approx 3/0.14$) times more efficient. The actual number is likely to be higher ($\times 40$) since there is already known to be about 7% amplitude contrast in images before the introduction of the energy filter (Toyoshima & Unwin, 1988*a*). For the more strongly scattering objects found in metallurgical rather than biological specimens, the advantage of the phase contrast mechanism is less.

Finally, there is the method of scanning differential phase contrast microscopy which offers the same advantages as bright field phase contrast microscopy but is used in scanning mode with an incident beam focused to a point. In practice, the detector performance is a limitation.

4. RELATIVE INFORMATION CONTENT OF PHASE CONTRAST COMPARED WITH HOLOGRAPHY AND DIFFRACTION

The type of phase contrast microscopy discussed above has many parallels with holography (Gabor, 1948), the only difference, apart from coherence requirements, being that in addition to the scattered and unscattered illuminating beams, holography* requires an additional, coherent and normally stronger, reference beam which passes by a different path to the recording medium (Lichte, 1991; Tonomura, 1992). Apart from the differences in the geometrical distribution of the contrast $\{\cos\chi$ instead of $\sin\chi$, where $\chi = 2\pi/\lambda[(\Delta F\theta^2/2) - (C_s\theta^4/4)]$ and λ = wavelength, ΔF = defocus, C_s = spherical aberration, θ = scattering angle}, its magnitude scales in the same way with scattering as phase contrast, and therefore offers all the same advantages.

Diffraction techniques are normally used on crystals where the discrete spots are easily measured in the diffraction plane above any surrounding background. It can be shown that the intensity of a sharp diffraction spot containing a certain number, N , of diffracted quanta will be measured with the same accuracy ($\sqrt{N}$) as would the amplitude (squared) of the corresponding Fourier component in the bright field phase contrast image that would result from interference of this scattered beam with the unscattered beam (Henderson, 1992). The diffraction pattern, if recorded at high enough spatial resolution, would therefore contain all the intensity information on Fourier components present in the image. It would lack only the information concerning the phases of the Fourier components of the image which are of course lost.

* Holography here refers to off-axis holography. In-line holography is identical to bright-field phase contrast microscopy with a somewhat larger defocus than normally used, which requires greater spatial coherence in the illumination.

Thus, for the same exposure, holography should be equal to normal phase contrast in performance, and diffraction methods inferior because of the loss of the information on the phases of the Fourier components of the image.

5. NEUTRONS

Just as visible light does little damage to biological molecules because photons of this energy are not absorbed but only scattered by most components in living cells, the same is true to a large extent of neutrons in the 1–10 Å wavelength range which have an energy $< kT$, and are termed cold or ultracold. Neutrons are therefore possible candidates for microscopy of cells and individual biological molecules. Elastically scattered neutrons and scattered light photons deposit negligible energy in the specimen, in each case very much less than kT , the thermal energy background. Even inelastic coherent and incoherent neutron scattering (as usually understood by condensed matter physicists), because of the low kinetic energy of neutrons of this wavelength, does no damage to proteins or nucleic acids. The problem lies mainly with two nuclear reactions which, although they have relatively small cross-sections, deposit substantial amounts of energy when they occur, causing radiation damage. There is thus a certain level of damage from γ -rays and the deuteron recoil produced by the low probability (n, γ) radiative capture reaction for $^1\text{H} \rightarrow ^2\text{H}$, as well as energy liberated and direct damage in the (n, p) reaction of $^{14}\text{N} \rightarrow ^{14}\text{C}$, which is estimated in Table 1. If fully deuterated, ^{15}N -labelled material is used this drops to a lower but still non-zero value (see Table 1). At a neutron wavelength of 1.8 Å, which is at the peak of the flux of thermal neutrons at the Institute Laue-Langevin, Grenoble (ILL), it can be seen that the damage is slightly higher than for electrons when applied to specimens with normal isotopes but $\sim 50 \times$ less for neutron irradiation of deuterated, ^{15}N -labelled molecules. If neutrons of wavelength 10^{-2} Å are considered (Table 1), the reduced inelastic cross-section for faster neutrons (the so-called ' $\frac{1}{v}$ law' (see Fig. 2) means that the energy deposited per useful elastic event would be lower than for 1.8 Å neutrons by a further factor of 180. It is therefore just possible that a combination of faster neutrons together with deuteration and ^{15}N -labelling would bring the energy deposited per useful elastic event into the range where single molecule microscopy without damage was viable.

Why not then develop neutron microscopy as a competitor to electron microscopy? Neutrons being uncharged may have further advantages. The problem is purely a practical one. Available neutron fluxes are not yet high enough for microscopy. Indeed, flux levels are so weak that the subject of neutron microscopy is rarely discussed (but see Steyerl *et al.* 1988; Steyerl, 1989). Regardless of the neutron wavelength selected as optimal from the point of view of radiation damage, the question of the flux level required is independent of neutron wavelength since elastic cross section does not vary with wavelength. For the purposes of this review, it is pertinent to ask by what factor are available fluxes too low. We might ask what would be needed in two different resolution ranges: 100 nm resolution which would be better than visible light, and 0.1 nm (1 Å)

which would allow atoms to be resolved within molecules and provide a possible competitor to electron microscopy.

6. THE FEASIBILITY OF NEUTRON MICROSCOPY

It can be seen (Fig. 2) that neutron elastic cross-sections are essentially 10^5 times less than electron cross-sections. For an acceptable signal to noise ratio it will be seen later that atomic resolution, phase contrast, electron images of single protein molecules of molecular weight 25000 would require incident doses of about $1000 \text{ el}/\text{\AA}^2$ in the absence of any radiation damage. A neutron phase contrast image with the same dose would have 300 times less contrast than an electron micrograph, scaling as the coherent scattering amplitude ($\sqrt{10^5}$), and would therefore require 10^5 times greater dose to provide the same signal to noise ratio, assuming that all problems of lenses and recording media could be solved. Thus, a dose of neutrons of $10^8/\text{\AA}^2$ would be needed. This must be compared with present fluxes at ILL, Grenoble of $10^{10} \text{ neutrons cm}^{-2} \text{ s}^{-1}$ before monochromatization. So, an exposure would require $(10^8/10^{10})10^{16} \text{ s} = 10^{14} \text{ s} = 3 \times 10^6 \text{ yr}$. In practice, it would be even longer than this if achromatic lenses were not available, and longer still if scanning modes were needed using only a Fresnel zone plate for illumination.

For a resolution of 100 nm, where we would be aiming to record images of 10^3 times larger features, the pixel size would be bigger by a factor of 10^6 and the required exposure correspondingly lower. Thus, it would seem that bright field phase contrast neutron microscopy at the 100 nm level might be attainable within a reasonable period if good lenses could be made. The exposure time might be a few years with present fluxes. On the other hand, if a scanning mode of lenseless imaging was necessary, exposure times would go up to several centuries because of the loss of the phase contrast mode.

In practical terms, for resolutions of 100 nm, neutrons of much longer wavelength would have to be considered because the construction of either mirror lenses or zone plates for neutrons in the 1 to $10 \text{\AA}$ wavelength region is virtually impossible (Clive Wilkinson, personal communication). Unfortunately, for these longer wavelength neutrons, fluxes are orders of magnitude lower than the $10^{10} \text{ neutrons cm}^{-2} \text{ s}^{-1}$ overall flux at ILL.

The overall conclusion is that atomic resolution neutron microscopy is impossibly far off, but that resolutions in the slightly sub-optical region might be worth further thought if some of the practical difficulties could be overcome.

7. ELECTRONS VERSUS X-RAYS

Table 1 and Figs 1 and 2 show some comparisons between electron and X-ray scattering cross-sections for typical organic materials. Table 1 and Fig. 2 include some data on neutrons for completeness. Note that the numbers refer to primary radiation damage as would be observed with specimens frozen at liquid nitrogen temperature. The situation with room temperature specimens is more complex (Henderson, 1990).

Table 1. Energy deposited in biological specimens per useful scattering event

	Electrons 80-500 keV	Neutrons			
		X-rays		1.8 Å	
		1.5 Å	30 Å	¹ H, ¹⁴ N labelled	² H, ¹⁶ N labelled (99 %)
Ratio † (inelastic/elastic) scattering events	3	10	10 ³ -10 ⁴	0.080	0.001 10 ⁻⁵
Mechanism of radiation damage	Secondary e ⁻ emission	Photoelectric e ⁻ emission		¹ H (n, γ) → ² H ¹⁴ N (n, p) → ¹⁴ C and recoil energy	Variety of lower cross-section reactions as well as residual ¹ H, ¹⁴ N
Energy deposited per inelastic event	20 eV	8 keV	400 eV	2 keV*	2 KeV*
Energy deposited per elastic event**	60 eV	80 keV	400 keV	160 eV	2 eV 0.02 eV
Energy deposited relative to electrons					
(inelastic)	1	400	20	100	100
(elastic)	1	1000	10000	2.5	0.003 0.0003

† Ratio of inelastic to elastic events for electrons and X-rays are well-known (see, for example, International Tables for crystallography, volume C). The values for neutrons were calculated from neutron scattering cross-sections given by Sears (1992) for the two nuclear reactions ¹H(n, γ) → ²H and ¹⁴N (n, p) → ¹⁴C, which have similar probabilities for typical protein composition. The black dots in Fig. 2 show where these points have been taken from in the overall curve describing the variation with wavelength.

* The 2 keV estimated energy deposited per neutron inelastic event is the average of the deuteron recoil energy in the first reaction (1.3 keV) and the ¹⁴C recoil energy in the second reaction (3 keV). The emitted γ-ray and proton in each case carries away a much larger energy but little of this would be deposited in the specimen. With X-ray and electron inelastic events, energy is deposited by the emitted electrons and secondary electrons respectively.

** Row 1 multiplied by Row 3.

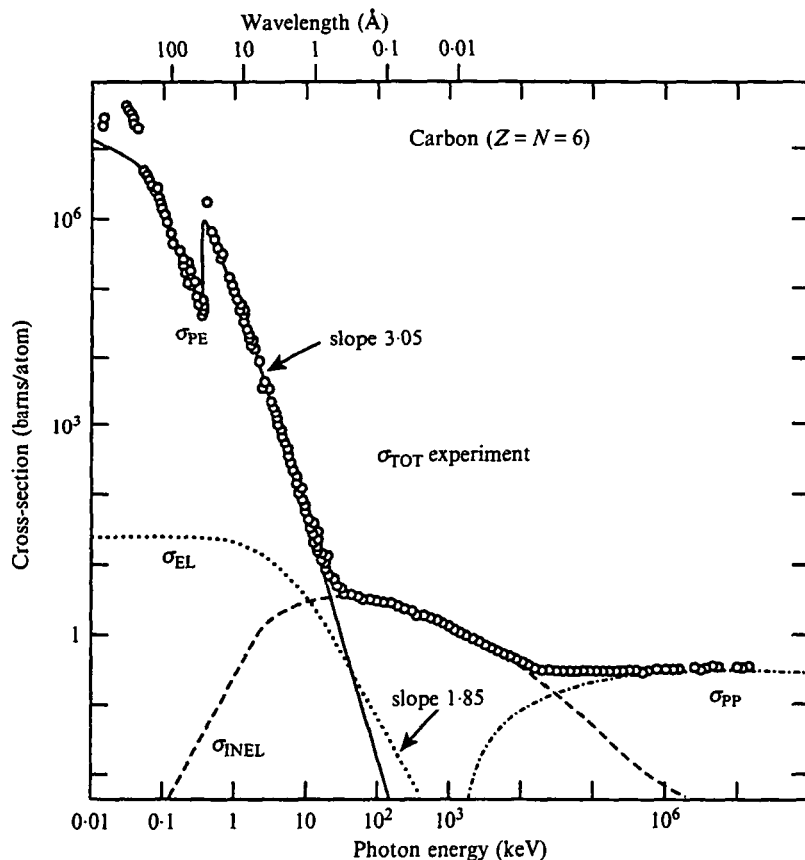


Fig. 1. Cross-section for photon scattering from carbon showing the contributions of photoelectric, elastic (Rayleigh), inelastic (Compton) and pair-production cross sections to the total cross sections. Also shown are the experimental data (open circles). Reproduced from Hubbell *et al.* (1980) with permission.

The key number to judge information content in phase contrast microscopy using X-rays or electrons is the amount of energy deposited by inelastic events per useful coherently-scattering, elastic event. It is this number that describes the extent of radiation damage that occurs for each unit of phase contrast information: the information in the scattered beams is fully recorded by interference with the unscattered beam in phase contrast microscopy. It can be seen that 1.5 Å X-rays are 1000 times worse and 30 Å X-rays 10000 times worse than electrons in this respect.

For inelastic events, usable in amplitude contrast modes, the best comparison shows X-rays (30 Å) producing 20 times more damage per inelastic event. In practice, the most favourable situation for X-ray microscopy occurs using absorption contrast in the water window between the carbon and oxygen K-absorption edges (23–44 Å). In this region, the absorption by protein molecules is 10 times greater than the absorption by an equivalent volume of water, whereas for electrons the contrast difference is only 0.4. This 25-fold contrast advantage

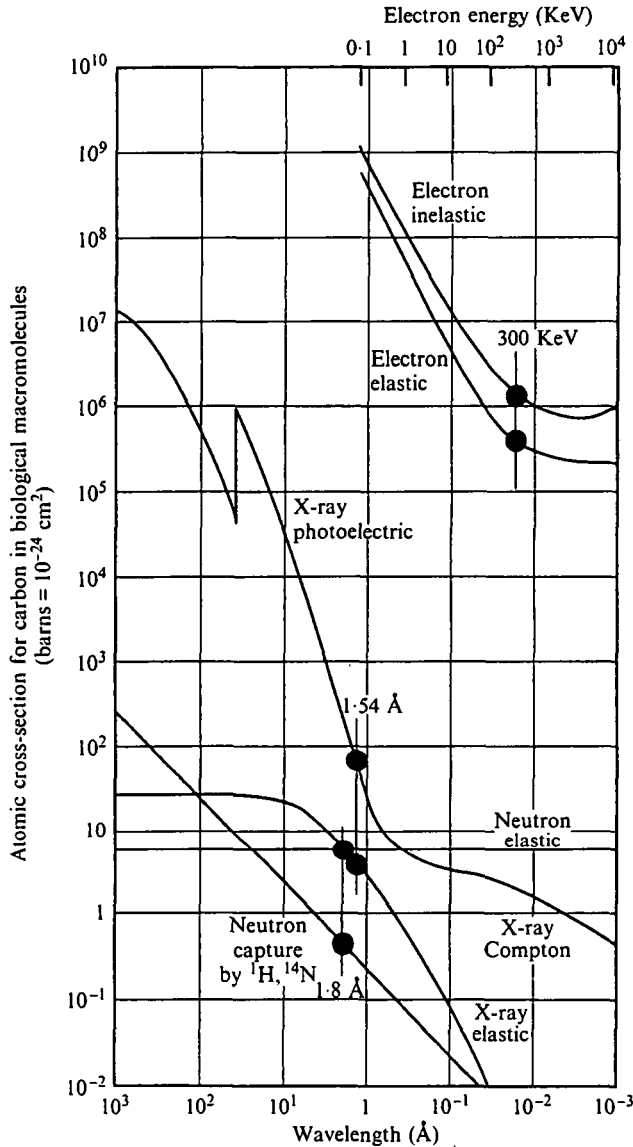


Fig. 2. Comparison of the atomic cross-sections for electrons, X-rays and neutrons as a function of wavelength of the particle or quantum. The elastic cross-sections all apply to carbon atoms, whereas the absorption and inelastic cross-sections represent the proportional value in a protein molecule scaled to be comparable to the carbon elastic cross-section. The coherent, elastic cross-sections are important for phase-contrast microscopy. The absorption and inelastic cross-sections contribute to radiation damage and arise from a variety of mechanisms. The circular black symbols show the positions on the wavelength scale used to derive the numbers in Table 1.

provides a considerable boost for X-ray microscopy in the water window over electron microscopy when modes of electron microscopy based on counting modes are considered. It has allowed Sayre *et al.* (1977) to state that 'In natural biological

specimens thicker than a critical thickness (100–3000 Å depending upon specimen composition), photons in mode X1 (bright field, transmission mode) can detect equidimensional features of any size at a lower level of dosage to the specimen than can electrons'.

However, if bright field, defocused, phase contrast electron microscopy is compared with X-ray microscopy (bright field, transmission mode) in the water window the comparison is not so clear. Table 1 shows a 7-fold lower damage per elastic electron scattered, compared with damage per inelastic, 30 Å X-ray absorbed [60 eV, compared with 400 eV]. This translates into a $\sqrt{7}$ -fold contrast advantage if the exposure is adjusted to give the same amount of damage. In addition, we have to consider the greater contrast efficiency of phase contrast compared with absorption. This varies depending on the specimen, but as discussed above, extrapolating from the work of Smith & Langmore (1992), TMV in ice is contrasted 40 times more efficiently when phase contrast is compared with absorption contrast, in this case based on removal of the energy loss electrons (similar to mode X1 in Sayre *et al.* 1977). Thus, for the same radiation damage we have a factor of 100 in favour of phase contrast electron microscopy to be balanced against the 25-fold higher contrast for X-ray absorption in the water window. Since we are comparing bright-field (and not dark field) imaging modes in both cases, the 25-fold contrast advantage is worth less than the 100-fold contrast advantage that is obtained directly after adjusting exposure for equal amounts of radiation damage.

Although the above discussion extrapolates from electron microscopy of only one specimen (TMV in ice), the relative advantage of X-rays over electrons, even in the water window, is therefore not proven and it seems likely that electrons produce more information when used to provide phase contrast, than X-rays of any wavelength including the region within the water window. However, it would be useful to recalculate this precisely using the formulation of Sayre *et al.* (1977).

One final example might help to clarify the relative power of phase contrast electron microscopy. Sayre *et al.* (1977) calculated that a dose of 10^9 rad was required to detect a 100 Å diam. protein molecule in water (of thickness from 100 Å to 10 μ m) using X-rays in the water window. At precisely this dose (2–5 el/Å²), it has been demonstrated that 20 Å thick DNA (Adrian *et al.* 1990), or single molecules of cytochrome c (Berriman, personal communication) which are approximately 30 Å diam., can be seen clearly using 100 keV bright field phase contrast electron microscopy. The calculations for electron microscopy given below in Table 2 confirm the detectability of protein molecules considerably smaller than 100 Å diam. Similar conclusions are reached in a recent paper by Sayre & Chapman (1994).

8. ELECTRON MICROSCOPY

There are already published structures of unstained biological molecules or assemblies which show it is possible to determine atomic resolution protein structure by electron microscopy. The work on PhoE (Jap *et al.* 1991), LHC-II

Table 2. Parameters in electron microscopy of single protein molecules or molecular assemblies*

Type of molecule	A†	B	C	D	E	F	G	
Large virus	300 M	900	25 000 000	141 371	0.0520	0.184×10^{-4}	0.420×10^{-3}	
Small virus	11 M	300	936 000	15 707	0.0173	0.552×10^{-4}	0.743×10^{-3}	
Ribosome	3.3 M	200	277 000	6981	0.0115	0.827×10^{-4}	0.910×10^{-3}	
	1.4 M	150	117 000	3926	0.0087	1.103×10^{-4}	1.050×10^{-3}	
Multimeric enzyme	420 K	100	35 000	1745	0.0058	1.654×10^{-4}	1.286×10^{-3}	
	180 K	75	14 600	981	0.0043	2.206×10^{-4}	1.485×10^{-3}	
Small protein	52 K	50	4330	436	0.0029	3.309×10^{-4}	1.819×10^{-3}	
Very small protein	18 K	35	1500	213	0.0020	4.727×10^{-4}	2.174×10^{-3}	
	7 K	25	540	109	0.00144	6.618×10^{-4}	2.572×10^{-3}	
Equation	(A1)	—	(A2)	(A3)	(A4)	(A5)	(A6)	
Relation to D	$0.418 \times D^2$	D	$0.0346 \times D^2$	$0.01745 \times D^2$	$5.7 \times 10^{-3} \times D$	$1.654 \times 10^{-4} \times D^{-1}$	$0.0128 \times D^{\frac{1}{2}}$	
Dependence on resolution d	—	—	—	$\alpha(1/d^2)$	—	—	—	
Type of molecule	H	I	J	K	L	M	N	O
Large virus	0.322	0.30	644	5.2	8.5	Yes	13	12 600
Small virus	0.186	0.30	124	4.8	7.7	Yes	40	12 600
Ribosome	0.152	0.30	68	4.7	7.5	Yes	60	12 600
	0.132	0.30	44	4.6	7.3	Yes	80	12 600
Multimeric enzyme	0.107	0.30	24	4.4	7.1	Possibly	120	12 600
	0.093	0.30	16	4.2	6.8	Possibly	160	12 600
	0.076	0.30	8.4	4.1	6.7	Possibly	240	12 600
Small protein	0.064	0.30	4.9	3.9	6.3	No	345	12 600
Very small protein	0.054	0.30	3.0	3.5	5.9	No	480	12 600
Equation	(A7)	(A8)	(A9)	(A13)	(A16)	(A17)	(A10)	(A11)
Relation to D	$0.0107 \times D^{\frac{1}{2}}$	—	$0.02388 \times D^{\frac{1}{2}}$	—	—	—	$12087 \times D^{-1}$	—
Dependence on resolution d	$\alpha(1/d)$	$\alpha(1/d)$	$\alpha(1/d)$	—	—	—	—	38 000/ d

* To simplify the presentation, it is assumed that the molecules are arranged in a closely-packed 2-dimensional crystal with a square unit cell as shown in Fig. 3. The formulae used to derive Table 2 are given in the Appendix.

† (A), Approximate molecular weight (Da); (B), D (Å); (C) N_e , number of carbon atom equivalents; (D), N_e , number of unique diffraction spots to resolution of $d = 3$ Å in projection; (E) f , fraction of electrons elastically scattered out to 3 Å resolution; (F), $\langle F_{\text{obs}} \rangle / F_0$; (G) $\langle F_{\text{obs}} \rangle / F_0$; (H), phase contrast = total image fractional contrast = signal; (I), fractional noise level in pixel of dimension $(d/2)^2 = 1.5 \text{ Å} \times 1.5 \text{ Å}$; (J), can single molecule be detected? How many times > noise? (K), multiple of sigma expected within unit cell at random; (L), multiple of sigma expected within entire volume of five parameter space at random; (M) can single molecule alignment be carried out in practice? (N) minimum number of images needed for structure with average Fourier component to be > 3 σ in projection; (O) total number of images in $3D \times [\pi D/d]$, De Rosier & Klug (1967).

(Kühlbrandt & Wang, 1991; Kühlbrandt *et al.* 1994) and bacteriorhodopsin (Henderson *et al.* 1990) has used the ability to average images of molecules in two-dimensional crystals to overcome the signal-to-noise limitation resulting from the dose limit imposed by radiation damage. In each of these cases, maps of the structure at atomic or near-atomic (3.4–6 Å) resolution have been obtained. Similar analyses of viruses embedded in amorphous ice (Jeng *et al.* 1989; Baker *et al.* 1991; Stewart *et al.* 1991), 50S subunits of ribosomes (Penczek *et al.* 1992) and tubular crystals of acetylcholine receptor (Toyoshima & Unwin, 1988*b*) or Ca²⁺-ATPase (Toyoshima *et al.* 1993) show that it is possible to carry out inter-particle averaging of either large enough particles (ribosomes), or smaller subunits arranged into structures with symmetries other than crystalline (e.g. helical or icosahedral). Although the published results on viruses and ribosomes display proven resolutions of only 9 Å (TMV) to 40 Å (ribosomes), and therefore are only slightly better than resolutions obtained with the use of heavy metal stains, the big difference is that the structural work in amorphous ice gives the structure of the molecules themselves and not that of a replica. With further improvements in imaging techniques, it is expected that it should be possible to determine the atomic structures of viruses and ribosomes by electron microscopy combined with digital image averaging techniques.

For the purposes of this review, I would like to ask the question: what is the smallest size of free-standing molecule whose structure can in principle be determined by phase-contrast electron microscopy? Given what has already been demonstrated in published work, this reduces to the question: what is the smallest size of molecule for which it is possible to determine from images of unstained molecules the five parameters needed to define accurately its orientation (three parameters) and position (two parameters) so that averaging can be performed? Perhaps it is useful to give first a simple intuitive explanation of why a large molecule is easier to align than a small molecule.

Provided a macromolecule is internally atomically ordered (i.e. no bending, twisting or squashing), the problem of alignment consists of the accurate determination of the two translational and three orientational parameters. The accuracy with which they must be determined does not change much with particle size, but there is a dependence on resolution. At a given resolution, the information to determine the five parameters comes from the image and the image of the macromolecule simply contains more information if the molecule is bigger because there is more scattering. Once alignment is achieved, the structure can then be calculated, so a sufficiently accurate alignment of the noisy images becomes the limitation.

Table 2 shows the results of calculations for hypothetical, spherical protein molecules or assemblies of various sizes from 25 to 1000 Å, composed of randomly positioned atoms. The model used is shown in Fig. 3 and the formulae used to derive Table 2 are given in the appendix. Two columns give the essential conclusions concerning whether the presence of a single molecule can be detected, and whether it is possible to determine its angular orientation, so that the application of averaging methods can be used to determine the structure. It can be

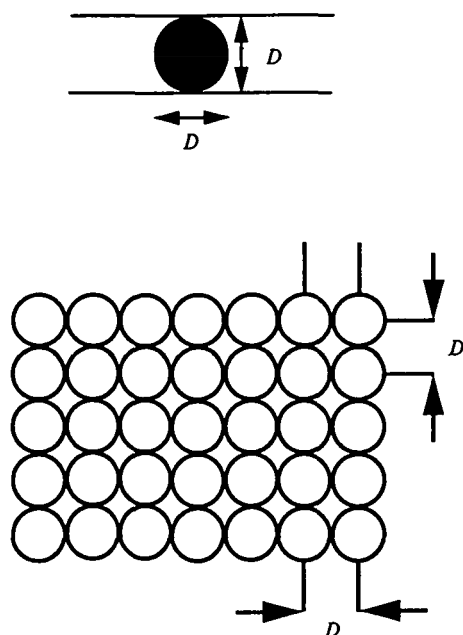


Fig. 3. Protein molecules of diameter D possibly embedded in a thin film of ice contain a certain number of atoms. A square crystalline array of the molecules is arranged as a two-dimensional crystal. This model is used to calculate the numbers given in Table 2.

seen that in principle, structures can be determined for any molecule of size larger than about 10^5 Da and in practice, with reasonable assumptions about image contrast, molecules larger than 10^6 Da are already within the accessible range. If specimens with more restricted orientations are available, then this size limit would be slightly lower, and, of course, it is the combined molecular weight of any higher symmetry aggregate that is important to consider. It therefore seems clear that there is great scope for further advance in high resolution electron microscopy for unstained biological structure determination. In practice, the remaining problems include the poor high resolution image contrast which is particularly bad for ice-embedded specimens. This is probably caused by blurring of the image due to beam-induced specimen movement and charging (Henderson, 1992).

9. WAVELENGTH AND ENERGY DEPENDENCE FOR ELECTRONS AND X-RAYS

The absorption (inelastic) and scattering (elastic) cross-sections for both electrons and X-rays vary steeply with energy and wavelength, whereas for thermal neutrons the elastic scattering cross section is constant (Fig. 2) and absorption strongly wavelength dependent. It might be thought therefore that considerable advantage could be gained by optimizing the energy of the incident illumination.

However, for X-rays and electrons, any change in the quality of data with energy turns out to be relatively small in practice.

In the case of X-rays, it can be seen from Fig. 1 for carbon that the photoelectric (absorption) cross-section excluding edge effects varies as $\lambda^{3.0}$ whereas the total elastic cross-section tends towards a $\lambda^{2.0}$ variation (Darwin, 1914) at the high energy end of the range. Therefore, at longer wavelength, relatively more X-ray quanta are absorbed than scattered. However, these quanta deposit less energy in the specimen, and so in terms of our criterion of energy deposited per useful elastic event, there should be little wavelength dependence. In diffraction, this has been discussed by Arndt (1984) who was particularly interested in the 0.5–1.5 Å range that is most important in high resolution X-ray diffraction from protein crystals. It should be noted that at higher energies (< 0.5 Å) the Compton scattering and pair production mechanisms begin to be important, so that there is only a limited range over which the energy deposition per useful elastic event is expected to be constant. At higher energies (shorter wavelengths) the situation would get worse.

It should also be noted that at longer wavelengths the total elastic cross-section (Fig. 1) becomes constant, implying that energy deposited per useful elastic event would become much greater (proportional to λ^2). However, this is not true for elastic scattering at a given desired resolution: the differential cross-section from scattering angles corresponding to a fixed resolution range, for example, from d_1 to d_2 , retains its λ^2 dependence (Darwin, 1914). The constant total elastic cross-section at longer wavelengths arises because of the loss of the ability to diffract beyond a resolution of $\lambda/2$ (the shrinkage of the Ewald sphere).

Finally, most protein crystallographers have found in practice that 0.9 Å X-rays give them more usable data than 1.5 Å X-rays within the time limit determined by radiation damage (Gonzalez *et al.* 1992). The reasons for this, still being investigated, may include the greater absorption of the diffracted beams at longer wavelength if the specimen is large or contains a strongly absorbing (high salt) mother liquor, or a possible heating effect of the powerful synchrotron beams that are used and therefore a greater effect of the same amount of energy deposition. Either of these explanations would still be consistent with the expectation of a very small energy dependence of the amount of useful diffraction or imaging data obtained for the same amount of radiation damage.

In the case of electrons, the cross sections for interaction get smaller at higher energy until damage from nuclear displacement becomes important when the inelastic cross-section increases again at about 1.5 MeV. This so-called 'knock-on' damage is thought to become measurable for hydrogen at about 5 keV and is therefore potentially important for organic or biological specimens, but in practice the effect is small when compared with 'bond' damage. The interaction cross-sections for elastic and inelastic events vary in a roughly parallel manner with energy, being inversely proportional to the electron velocity which is 0.55c at 100 keV, 0.70c at 200 keV and 0.86c at 500 keV. Thus, above 200 keV as the electron speed approaches c , the situation changes only slightly. This can be seen in Fig. 2 as the flattening out of the cross-sections at high energy. Below 100 keV,

however, both cross-sections vary strongly with energy tending towards a λ^2 dependence at low energy. However, since inelastic events deposit an average of about 20 eV of energy (Langmore & Smith, 1992) by interaction with plasmons and outer shell electrons with a much smaller cross-section for the more energetic inner shell excitations, the damage per inelastic event is thought to be more or less independent of electron energy. Therefore, the amount of damage (energy deposited) per useful elastic event should also be relatively independent of the energy of the incident electron beam.

The two most important considerations in the selection of the voltage for electron microscopy of biological specimens are the need (a) to minimize the influence of dynamical (multiple) scattering (Glaeser & Ceska, 1989) which reduces the information content and interpretability of the data and (b) to minimize the influence of specimen charging on the path of the electron beam and therefore on the amount of image blurring which occurs in low dose images. Both of these goals are best achieved by selection of the highest operating voltage available, though this can be an expensive choice. This also results in shorter wavelength and a flatter Ewald sphere, which has additional advantages in extending the region of faithful image formation to include thicker objects.

As a postscript, it could be added that high energy proton or other heavier charged particle microscopy would be expected to have roughly the same ratio of elastic to inelastic cross-sections, a similar energy loss per inelastic event, and therefore to behave in a generally similar way to electrons. However, because of the greater mass of the proton, the cross-sections for knock-on damage would be higher, so they would be expected to produce slightly more damage per useful elastic event, and therefore be less useful than electrons for high resolution biological microscopy.

10. CONCLUSION

Electron microscopy offers great scope for immediate improvements in capability by addressing the practical problems of specimen movement and charging of specimens embedded in amorphous ice (Henderson, 1992). Images of molecules or molecular assemblies of molecular weights of $\sim 10^5$ and above should contain enough information to determine the orientation and alignment of the particle being observed, so that subsequent image averaging methods may be used to determine the atomic structure. The number of particle images which must be averaged is at least 10000 (Table 2), making a large but manageable computing task. Surprisingly, the number of independent images required is independent of the particle size, being greater than 10000 in all cases. This drops to about 4000 at 10 Å resolution, and about 2000 at 20 Å resolution (see Appendix). Of course, in practice, the quality of the data is likely to be less than perfect, so it will be necessary to average considerably more than this to produce satisfactory maps of the structures. In the three-dimensional density map of bacteriorhodopsin (Henderson *et al.* 1990), the images of approximately 5 million molecules were averaged to produce recognizable amino acid side chains. In a similar analysis of

the structure of LHCII (Kühlbrandt *et al.* 1994), 2.6 million molecules were averaged.

X-ray microscopy using the phase contrast method is still limited in resolution by the present state of the art of zone plate construction. Even then (Table 1) the amount of damage per useful elastic event for thin specimens is several hundred times worse than for electrons but for thicker specimens the balance tilts back in favour of X-rays. The advent of electron energy-filtering as a practical tool extends the usable thickness for specimens being studied by electron microscopy up to thicknesses of several hundred nanometres. Given a mean free path for penetration for 200 keV electrons of about 800 Å, this means that for general purpose microscopy the cross-over into the region where X-rays offer a superior performance will occur at around 5000 Å in thickness – where the number of elastic electrons has dwindled due to inelastic collisions to the level where the energy deposited per useful elastic event is equal to that obtained for X-rays. Extrapolating from the calculations given in Table 2 for electrons, where 10000 particles are needed for a structure determination in 3D, the comparable number for X-ray analysis using 1.5 Å X-rays would be higher by a factor of 10^3 (Table 1), giving a requirement for 10^7 molecules, or a crystal with all three dimensions around 5 µm for an average unit cell. Thus, the determination by electron microscopy of the structure of a macromolecule from 10000 single particles is just as possible as solving an X-ray structure from 5 µm crystals.

There are at least two areas where X-ray microscopy offers compensating advantages (Jacobsen *et al.* 1991; Kirz *et al.* 1992). First, in the water window, the 25-fold higher contrast will offset the disadvantage of much higher radiation damage, so that absorption microscopy which depends on damaging events for its contrast, should provide useful images in the 500 Å resolution range. Secondly, the tuning of the X-ray wavelength to take advantage of chemical absorption edges offers the possibility of chemical resolution for example in DNA versus protein versus lipid in biological structures. It should be noted that similar analyses are possible using various modes of electron imaging but the required electron doses are much higher than discussed above for phase contrast electron microscopy.

Finally, a promising but unfortunately distant possibility is for high resolution neutron microscopy using deuterated, ^{15}N -labelled specimens. If the brightness of sources could be increased 10^6 -fold over those presently available and shorter wavelengths used, neutrons might be very valuable for biological microscopy and diffraction. No doubt, there would be many practical problems such as lens and detector limitations, but work might be expected to progress steadily from initial efforts at low (sub-micron) resolution using non-isotope labelled material in D_2O at longer wavelengths.

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13. APPENDIX: FORMULAE FOR TABLE 2

(A) Molecular weight (M_w) = $\frac{4}{3}\pi\left[\frac{D}{2}\right]^3\rho$, (A 1)

where ρ is the protein density $\approx 0.8 \text{ Da}/\text{\AA}^3$

(B) The fundamental variable is the diameter, D , of a spherical protein ($\text{\AA}$).

(C) Number of carbon atom equivalents, using 9.1 carbon equivalents per amino acid of molecular weight = 110:

$$N_c = \frac{M_w \times 9.1}{110} = \frac{4}{3}\pi\left[\frac{D}{2}\right]^3\rho\frac{9.1}{110}. \quad (\text{A } 2)$$

(D) Number of unique diffraction spots to resolution d , in projection

$$N_s = \frac{\pi}{2}\left[\frac{D}{d}\right]^2. \quad (\text{A } 3)$$

(E) Fraction, f , of electrons elastically scattered out to $3 \text{\AA}$ resolution is approx. one third of total elastically scattered electrons (see for example Langmore & Smith (1992) equation 4).

$$f = \frac{\sigma_e(c)}{3}\left[\frac{N_c}{D^2}\right],$$

where $\sigma_e(c) = 0.005 \text{\AA}^2$ is the elastic cross-section for carbon (Reimer, 1989)

$$= \frac{\sigma_e(c)}{3}\frac{4}{3}\pi\left[\frac{D}{2}\right]^3\rho\frac{9.1}{110}\left[\frac{1}{D^2}\right]. \quad (\text{A } 4)$$

(F) The mean intensity $\langle I_{\text{obs}} \rangle$ and (G) mean amplitude $\langle F_{\text{obs}} \rangle$ of the diffracted beams as a fraction of the incident beam are given by the following expressions.

$$\frac{\langle I_{\text{obs}} \rangle}{I_0} = \frac{f}{2N_s} = \frac{\sigma_e(c)}{3}\frac{4}{3}\pi\left[\frac{D}{2}\right]^3\rho\frac{9.1}{110}\left[\frac{1}{D^2}\right]\frac{1}{\pi}\left[\frac{3}{D}\right]^2 \quad (\text{A } 5)$$

$$= \frac{\sigma_e}{30D}$$

$$\frac{\langle F_{\text{obs}} \rangle}{F_0} = \sqrt{\frac{\langle I_{\text{obs}} \rangle}{I_0}}$$

$$= \left[\frac{\sigma_e}{30D}\right]^{\frac{1}{2}}. \quad (\text{A } 6)$$

Note that $\langle I_{\text{obs}} \rangle / I_0$ and $\langle F_{\text{obs}} \rangle / F_0$ are independent of resolution, d . The substitution of N_s in equation (A 5) specifically involves the number of spots at $3 \text{\AA}$

resolution since this is the resolution used to calculate the fraction of elastically scattered electrons in equation (A 4). Thus $d = 3 \text{ \AA}$ in equation (A 5) and the resulting formulae (A 5) and (A 6) are both independent of resolution.

(H) To calculate the mean phase contrast in the image, the contrast from a single Fourier component (two Friedel-related diffraction spots (Henderson & Glaeser, 1985)) is multiplied by the square root of the number of independent spots since each Fourier component is independent. Therefore, mean phase contrast in image as fraction of mean image density,

$$\begin{aligned} \text{Signal} &= 2 \frac{\langle F_{\text{obs}} \rangle}{F_0} \sqrt{N_s} \\ &= 2 \left[\frac{\sigma_e \pi D^2}{30 D^2 d^2} \right] \\ &= \left[\frac{\sigma_e \pi D}{15 d^2} \right]^{\frac{1}{2}}. \end{aligned} \quad (\text{A } 7)$$

(I) Fractional noise level in a single $[(d/2) \times (d/2)]$ pixel, at dose of $N_e/\text{\AA}^2$ electrons/ $\text{\AA}^2$

$$\begin{aligned} \text{Noise} &= \frac{\sqrt{N_e d^2/4}}{N_e d^2/4} \\ &= \frac{1}{\sqrt{N_e d^2/4}}. \end{aligned} \quad (\text{A } 8)$$

(J) Can a single molecule be detected? How many times sigma?

$$X_{\text{sig}} = \frac{\text{Signal}}{\text{Noise}} \times \sqrt{N_{\text{pix}}},$$

where N_{pix} is the number of pixels in one unit cell and corresponds to the area

$$\text{being searched: } N_{\text{pix}} = \frac{D^2}{d^2/4}$$

$$= \frac{\text{Signal}}{\text{Noise}} \times \frac{D}{d/2},$$

where Signal and Noise are previous two formulae (A 7) and (A 8)

$$\begin{aligned} &= \left[\frac{\sigma_e \pi D}{15 d^2} \right]^{\frac{1}{2}} \sqrt{N_e d^2/4} \frac{D}{d/2} \\ &= \left[\frac{\sigma_e \pi N_e D^3}{15 d^2} \right]^{\frac{1}{2}}. \end{aligned} \quad (\text{A } 9)$$

(N) Minimum number of images needed for projection structure with average Fourier component to be $> 3\sigma$, i.e. greater than 10 electrons in each spot on

average (in projection). The statistics of each Fourier component of the electron diffraction pattern is the same as that of the corresponding Fourier component in the bright-field phase contrast image (Henderson, 1992).

$$\begin{aligned}
 N_{\text{inproj}} &= \frac{10}{\frac{\langle I_{\text{obs}} \rangle}{I_0} D^2 N_e} \\
 &= \frac{10}{DN_e} \frac{30}{\sigma_e}
 \end{aligned} \tag{A 10}$$

(O) Therefore, total number of images needed for 3D structure

$$\begin{aligned}
 &= N_{\text{inproj}} \frac{\pi D}{d} \\
 &= \frac{10}{DN_e} \frac{30}{\sigma_e} \frac{\pi D}{d} \\
 &= \frac{300\pi}{N_e \sigma_e d} \\
 &= \frac{300 \cdot \pi}{500053} \left. \vphantom{\frac{300 \cdot \pi}{500053}} \right\} \text{ to } 3 \text{ \AA} \\
 &= 12600
 \end{aligned} \tag{A 11}$$

At lower resolutions, this would be a smaller number

$$\begin{aligned}
 &\approx 3800 \text{ to } 10 \text{ \AA} \\
 &\approx 1900 \text{ to } 20 \text{ \AA}
 \end{aligned}$$

We now calculate *the statistics of frequency of occurrence* by chance of such signal-to-noise. Strictly speaking independent cross-correlation peaks occur only once per pixel, but it is possible to adopt more or less stringent criteria for the accuracy required in sampling the cross-correlation function. The criterion adopted here is to consider the cross-correlation function more finely than once per pixel so that the accuracy in estimation of position and orientation is sufficient for the required resolution. Thus, in a search of a cross-correlation map, the number of cross-correlations which must be calculated to locate a molecule in a known orientation to within ± 0.1 pixel is given by

$$\begin{aligned}
 \text{Number of positions to examine} &= \left(\frac{2D}{0.2d} \right)^2 \\
 &= \left(\frac{10D}{d} \right)^2.
 \end{aligned} \tag{A 12}$$

The size of any real correlation peak must therefore be larger than that observable by chance in the Normal distribution of noise exhibited by the above number of

cross-correlation functions examined. For this number of points, the real peak measured in multiples of the standard deviation (**K**) must therefore be larger than x , where $\operatorname{erfc}(x/\sqrt{2}) = 1/\left(\frac{10D}{d}\right)^2$ where $\operatorname{erfc} t = 1/\sqrt{\pi} \int_t^\infty e^{-z^2} dz \approx \frac{0.56}{t} e^{-t^2}$ in the limit of large values of t (Abramowitz & Stegun, 1965, p. 316).

$$\text{Thus } x = \sqrt{2} \operatorname{erfc}^{-1}[(d/D)^2/100]. \quad (\text{A } 13)$$

To determine the three orientational angles as well as the position, the number of correlation coefficients which should be examined is the number of positions multiplied by the number of orientations.

$$\begin{aligned} \text{Number of orientations} &= \left(\frac{2\pi D}{0.2d}\right)^3 \\ &= \left(\frac{30D}{d}\right)^3. \end{aligned} \quad (\text{A } 14)$$

Combining equations (A 12) and (A 14), total number of cross-correlation coefficients to examine

$$\begin{aligned} &= \left(\frac{30D}{d}\right)^3 \left(\frac{10D}{d}\right)^2 \\ &= 3 \times 10^6 \left(\frac{D}{d}\right)^5. \end{aligned} \quad (\text{A } 15)$$

(**L**) Therefore the multiple of sigma expected within the entire volume of five parameter space to be examined

$$= \sqrt{2} \operatorname{erfc}^{-1} \left[\left(\frac{d}{D} \right)^5 / 3 \times 10^6 \right] \quad (\text{A } 16)$$

Note that the factor 3×10^6 is a very rough estimate which does not strongly affect the result of the calculation given in equation (A 16).

The question of whether it is possible to determine the position and orientation of a molecule from its image can now be decided by asking whether the calculation of the expected signal-to-noise ratio (equation (A 9)) exceeds that expected by chance (equation (A 16)). This is shown in Table 2 (column **M**) with the comment yes, no or 'possibly'. The 'possible' category arises because, in practice, the image contrast obtained is never as good as expected theoretically from the above calculations. Suppose image contrast is reduced from the theoretical value to a fraction C , then instead of the results of equation (A 9) being used directly in a comparison with equation (16), the reduced contrast will reduce the value of equations (A 7) and (A 9) by C . The condition for success will then be

$$0.024 D^{3/2} C > 7$$

$$C D^{3/2} > 300.$$

Using equation (A 1), this means

$$M_w > \frac{4}{3} \pi \rho 300^2 / 8C^2$$

$$> 38000 / C^2 \quad (\text{A } 17)$$

Thus, orientation is possible using perfect images for molecules of molecular weight greater than 38 kDa, but this increases dramatically as image quality decreases. For example,

for	$C = 0.4,$	$M_w > 230 \text{ KDa}$
for	$C = 0.1,$	$M_w > 4 \text{ Mda}$
and for	$C = 0.05,$	$M_w > 16 \text{ Mda}$

At present, images have a resolution-dependent fade-out of image contrast which means that C varies from 0.8 to 0.05 or less depending on the resolution examined (see Henderson, 1992).

Finally, we might ask how the results of the above calculations, given in Table 2 for a resolution of $d = 3 \text{ \AA}$ are affected if lower resolution images are considered. Those columns of Table 2 that are affected are indicated at the bottom of the table.

Equation (A 9) is proportional to $1/d$ and the numbers calculated by equations (A 13) and (A 16) also get slightly smaller as resolution d is decreased from $1/3 \text{ \AA}^{-1}$ to $1/10$ or $1/20 \text{ \AA}^{-1}$, because the position and orientation do not need to be determined so accurately at lower resolution. The two effects do not quite cancel out, and for this random atom calculation based on a crystal lattice (Fig. 3) the minimum size limit that can be dealt with actually increases at lower resolution. However, for single molecules embedded in ice and not part of a crystal, the value of $\langle F_{\text{obs}} \rangle / F_0$ gets larger than that predicted by the random atom model at low resolution, and this is more than likely to offset the loss of contrast due to low resolution in this calculation. In practice, therefore, the conclusions given above in equation (A 17) are essentially the same. The molecular weight limit is 38 kDa and this increases as image contrast, C , decreases in proportion to $1/C^2$.